CS 4632 Modeling and Simulation
Milestone 3 Report

Jordan Reid

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1 Implementation Summary

1.1 Overview

The restaurant simulation system was fully implemented using a discrete-event simulation approach. The system models customer flow through a restaurant including arrival, waiting, seating, cooking, and completion.

1.2 Core Features

- Customer lifecycle management (arrival, waiting, service, completion, abandonment)
- Event-driven simulation using a priority queue
- Server and cook resource allocation
- Queue handling for waiting customers
- Customer patience and abandonment logic
- Inventory and food cost tracking
- Customer satisfaction calculation

1.3 Architecture

The system consists of:

- SimulationEngine (core logic)
- Entities (Customer, Server, Cook)
- MetricsCollector (statistics tracking)
- InventoryModel (cost tracking)
- Event system (event scheduling and execution)

1.4 Parameterization

Key configurable parameters include:

- Number of servers and cooks
- Number of tables
- Arrival rates
- Simulation duration
- Capacity thresholds

1.5 Scope Updates

- Added CSV logging system with event-level tracking
- Implemented customer ID tracking for all events
- Added end-of-run summary export to CSV
- Improved server load balancing logic

2 Execution Documentation

2.1 Run Summary Table

c = cook count, s = server count, t = table count. Success determined by customer satisfaction levels, abandonment rates, and food cost to waste ratio. The console output for each run can be seen in the "Console Output Results" section.

Run ID	Purpose	Key Parameters	Customers	Success
1	Default	c3 s6 t50	119	False
2	Default (alt)	c3 s6 t50	116	True
3	Cook Increase	c4 s6 t50	149	True
4	Cook Decrease	c2 s6 t50	79	False
5	Server Increase	c3 s7 t50	121	True
6	Server Decrease (run1)	c3 s5 t50	103	True
7	Server Decrease (run2)	c3 s5 t50	116	False
8	Server Decrease (run3)	c3 s5 t50	129	False
9	Cook Decrease, Server Increase	c2 s7 t50	106	False
10	Cook Increase, Server Decrease	c4 s5 t50	121	True
11	Tables Increased	c3 s6 t60	116	False
12	Tables Decreased	c3 s6 t30	119	True
13	Tables Decreased Further	c3 s6 t20	130	False

2.2 Execution Environment

- Language: Java
- Simulation Type: Discrete Event Simulation
- Output Format: CSV logs and console summaries

2.3 Issues Encountered

- Log file path issues (resolved using relative paths)
- Customer ID mismatch (fixed by tying events to event objects)
- CSV formatting improvements for analysis

3 Data Collection Overview

3.1 Metrics Collected

Event-Level Data

- Timestamp
- Event type
- Customer ID
- Event details (server/cook assignments, satisfaction)

Summary Metrics

- Total customers
- Abandonment rate
- Average satisfaction
- Food cost ratio
- Success criteria

3.2 Data Format

Each run generates a CSV file:

```
Time,Event,CustomerID,Details
0.0,ARRIVAL,1,Entered restaurant
...
```

Followed by:

```
===== Simulation Summary =====
Total Customers,XXX
Abandonment Rate,X.X
...
```

4 Preliminary Results

4.1 Summary Table

4.2 Key Observations

- Increasing cooks significantly improves satisfaction
- Reducing cooks leads to high abandonment and poor performance
- Server count affects overall flow but is less critical than cooking capacity
- Table limits create bottlenecks under high demand
- System performance depends heavily on balanced resource allocation

4.3 Performance Insights

- Best performance achieved with higher cook availability
- Worst performance observed when cooking resources are limited
- Customer satisfaction strongly correlates with service time
- Abandonment increases under resource constraints

5 Console Output Results

The following shows the summary console output from each simulated run. Some starting parameters had multiple runs as the expected outcome of failing to meet the success criteria didn't happen when the starting staff had been reduced from the default. The reason behind the unexpected outcome stems from the slight randomization in customer quantity and the variations in service time.

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 119
Abandonment Rate: 0.0
Average Satisfaction: 62.137460522621524
Food Cost Ratio: 0.24
Meets Success Criteria: false
```

Figure 1: Default run: cooks 3, servers 6, tables 50

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 116
Abandonment Rate: 0.0
Average Satisfaction: 75.23485573990033
Food Cost Ratio: 0.23399999999999999
Meets Success Criteria: true
```

Figure 2: Default run 2, representing randomization elements in code

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 149
Abandonment Rate: 0.0
Average Satisfaction: 99.04986360773403
Food Cost Ratio: 0.3
Meets Success Criteria: true
```

Figure 3: Cooks increased to 4

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 79
Abandonment Rate: 0.0
Average Satisfaction: 59.93703769560422
Food Cost Ratio: 0.16
Meets Success Criteria: false
```

Figure 4: Cooks decreased to 2

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 121
Abandonment Rate: 0.0
Average Satisfaction: 73.68970439402258
Food Cost Ratio: 0.244
Meets Success Criteria: true
```

Figure 5: Servers increased to 7

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 103
Abandonment Rate: 0.0
Average Satisfaction: 90.71033767437855
Food Cost Ratio: 0.20800000000000002
Meets Success Criteria: true
```

Figure 6: Servers decreased to 5, run 1, unexpected outcome scenario

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 116
Abandonment Rate: 0.0
Average Satisfaction: 63.91947269953118
Food Cost Ratio: 0.23399999999999999
Meets Success Criteria: false
```

Figure 7: Servers decreased to 5, run 2, more expected outcome

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 129
Abandonment Rate: 0.08527131782945736
Average Satisfaction: 47.74767566382747
Food Cost Ratio: 0.23800000000000002
Meets Success Criteria: false
```

Figure 8: Servers decreased to 5, run 3, more expected outcome

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 106
Abandonment Rate: 0.27358490566037735
Average Satisfaction: 21.924573737929908
Food Cost Ratio: 0.156
Meets Success Criteria: false
```

Figure 9: Cooks decreased to 2 with servers increased to 7

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 121
Abandonment Rate: 0.0
Average Satisfaction: 100.0
Food Cost Ratio: 0.244
Meets Success Criteria: true
```

Figure 10: Cooks increased to 4 with server decreased to 5

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 116
Abandonment Rate: 0.0
Average Satisfaction: 66.01752868123212
Food Cost Ratio: 0.23399999999999999
Meets Success Criteria: false
```

Figure 11: Max tables increased to 60

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 119
Abandonment Rate: 0.0
Average Satisfaction: 75.84897536565086
Food Cost Ratio: 0.24
Meets Success Criteria: true
```

Figure 12: Max tables decreased to 30

```
Log file created at: D:\Study\School\Courses\Fall 2026\Modeling and Simulation\Project\Project_Code_Files\RestaurantSimulation\simulation_log.csv
===== Simulation Summary =====
Total Customers: 130
Abandonment Rate: 0.07692307692307693
Average Satisfaction: 61.400675178167106
Food Cost Ratio: 0.242
Meets Success Criteria: false
```

Figure 13: Max tables decreased to 20(extreme case)

6 Conclusion

The simulation successfully models a basic restaurant system with dynamic resource allocation and customer behavior. The results demonstrate how different parameters impact system performance, particularly highlighting the importance of cooking time and available employees.